

GNN Æ— MDT: what was done, why, what it showed, and what comes next

Nicolas Bigeard Æ” report for A. Kalogeratos, 2026-07-24. Full evidence: GRAPH_MDT_RESEARCH_LEDGER.md/.pdf, results/graph_mdt/ (600 result rows, raw), GNN_RESEARCH_LEDGER.md (first study).

1. Starting point and why a second study existed at all

The first GNN study (experiments/gnn_mdt/, ledger G1-G6) concluded that GNNs add nothing to MDT: distillation fidelity, graph-specific out-of-sample extension, graph autoencoders, DGI, learned view weights and compute were all closed as dead ends after a 6-dataset screen and a 6-dataset untouched validation (540 rows). Its one surviving result: parametric out-of-sample extension beats Nystr m (+0.048 pooled test AMI), but the effect belongs to the MLP, not to message passing.

Before presenting that as "we should not do GNN", I audited it and found the conclusion was not defensible in that form, for five reasons:

1. **Every graph was a kNN re-derivation of the features.** The graph could not, by construction, contain information the feature encoder lacked. A GNN's null result on such graphs is near-tautological.
2. **Datasets were subsampled to 800 train / 400 test nodes**, which shreds graph structure and puts the comparison in the small-sample regime where spectral methods are strongest.
3. **Cora and CiteSeer showed no signal for any method** Æ” a red flag for the pipeline, not for GNNs: these are the canonical graph benchmarks.
4. **Several verdict labels contradicted the ledger's own decision rule** (prose said "dead end" where the interval did not exclude the +0.02 minimum useful effect).
5. Fixed untuned configs, a single metric, three objectives Æ” thin coverage for a claim about "GNNs" as a class.

The honest options were: present a much narrower claim, or run one decisive study that removes every objection. We chose the second.

2. Design of the second study and the reasoning behind each choice

Requirements set in advance: independent graph information, intact large graphs, multiple GNN families and objectives, additional metrics, more independent datasets Æ” and three additions the requirement list missed:

- **A graph-independence screen as a gate, not an excuse.** For every dataset, measure whether the natural graph is recoverable from a feature-kNN graph (edge Jaccard at matched density) and whether it carries label-relevant structure (edge homophily + Platonov's adjusted homophily). Datasets whose graphs are feature re-derivations cannot test the hypothesis and are excluded before any model runs.

- **The same-inputs principle.** The fair opponent for a GNN is not feature-only MDT but **MDT given the same graphs as views** (adjacency $\hat{+}$ row-stochastic transition $\hat{+}$ one more view in the trajectory product). If spectral consensus with identical inputs matches the GNN, the GNN has no case. This single decision shaped everything downstream.
- **Pre-registration with code-enforced verdicts.** Thresholds (+0.02 AMI minimum useful effect, 95% t-intervals clustered by dataset, no verdict under 5 datasets), label-free model selection, and the rule that verdict labels are emitted only by `verdicts.py` $\hat{=}$ never written by hand. Every deviation is a timestamped amendment recorded before the rows it affects. This directly fixes flaw 4 of the first study.

Literature came first (zero compute): published multiplex clustering numbers for ACM/DBLP/IMDB were collected from O2MAC, DMGI, HDMI, MvAGC, MCGC, DualLGR, BTGF, InfoMGF, BMGC and cross-checked. Two findings shaped the study: (a) **two incompatible dataset lineages share the same names** (DBLP-4057-authors vs DBLP-7907-papers; IMDB-4780 vs IMDB-3550) $\hat{=}$ copying numbers across lineages, which several papers do, is meaningless; (b) cross-paper re-runs of the same method diverge by up to 30 NMI points, so placement targets a published range, never a single number.

3. What was built

`experiments/graph_mdt/` $\hat{=}$ self-contained suite, no torch-geometric:

- `datasets.py` $\hat{=}$ unified loaders (DMGI pickles, multi-view .mat, heterophilous npz) $\hat{+}$ (features, adjacency views, labels), cached npz. Nine datasets: acm, dblp, imdb (multiplex metapath graphs), cora, citeseer (intact citation graphs), amazon_ratings, minesweeper, roman_empire (heterophily controls, Platonov 2023), amazon (held-out validation). No subsampling anywhere.
- `independence.py` $\hat{=}$ the screen (step 2).
- `placement.py` $\hat{=}$ full-graph protocol runs matching the literature (all nodes, KMeans on embedding, ACC/macro-F1/NMI/ARI/AMI, 5 seeds).
- `methods.py` $\hat{=}$ implicit sparse MDT (LinearOperator SVD, works at 24k nodes without dense n^2), consensus and trajectory-selection fusion, 2-layer GCN/SAGE/MLP encoders with hand-rolled sparse message passing, GAE and BGRL objectives, GNN $\hat{+}$ MLP distillation, per-node view gating, A^2 two-hop views, modularity-based label-free selection.
- `grid.py / verdicts.py` $\hat{=}$ resume-safe runner; mechanical verdicts.

4. Results, round by round, with causes

4.1 Independence screen (13 graph views)

All natural views passed the Jaccard gate (max 0.11): none is a feature re-derivation $\hat{=}$ the exact property the first study lacked. The per-view homophily table later predicted method behaviour: ACM PAP (0.73 adjusted homophily vs 0.56 for feature-kNN) and DBLP PAP/PPrefP carry genuine extra signal; IMDB MAM sits below the feature graph (0.17 vs 0.24); DBLP PATAP (0.009) is noise; roman_empire's graph is anti-correlated with labels ($\hat{=}$ 0.047) while its features are strong (0.585).

Two harness bugs surfaced and were fixed here, both relevant to the lab beyond this study:

- **Gaussian-euclidean kNN collapses on bag-of-words features:** measured ACM feature-graph homophily 0.36 (euclidean) vs 0.71 (cosine). This is distance concentration in high-dimensional sparse data. It also retroactively explains part of the first study's dead Cora/CiteSeer.
- **Silhouette-based trajectory selection prefers collapsed embeddings.** On ACM, consensus reached NMI 0.63 while silhouette-selected trajectories returned 0.004: a noise-view-dominated diffusion smears all points into one blob, KMeans splits the blob into tight fake clusters, and silhouette rewards exactly that geometry. Selection now uses Newman modularity of the KMeans partition on the union input graph — still label-free, and degeneracy-proof because a collapsed partition has no modularity. The lab's select_trajectory uses the silhouette criterion and inherits this failure mode.

4.2 Placement against the published record (36 rows)

Full-graph NMI at dim 32, all label-free, zero training:

dataset	feat only	graph only	all views	published range
ACM	0.510	0.003	0.631	0.69–0.76
DBLP-7907	0.062	0.400	0.590	0.41–0.62 (HDMI 0.582)
IMDB-3550	0.148	0.005	0.002	0.196–0.198

Interpretation: on DBLP, MDT consensus with graph views **beats HDMI and every pre-2024 published method** and trails only BTGF (2024); on ACM it lands 0.06 under O2MAC. Fusion genuinely works where the screen said the graphs carry signal (+0.19 over the best single arm on DBLP) and fails exactly where the screen said the graph is weaker than the features (IMDB). Caveat stated up front: DMGI-lineage papers report test-split NMI while ours is full-graph; a paper version needs protocol-aligned reruns.

4.3 Main grid (440 rows: 11 arms – 8 datasets – 5 seeds)

Arms: MDT consensus / selection / feat-only / graph-only; GCN and SAGE backbones – GAE and BGRL objectives on the uniformly fused, top-64-capped adjacency; MLP+SSL and raw-KMeans feature references; GNN+MLP distillation. Uniform settings frozen in the ledger (dim 32, hidden 128, 150 epochs, modularity checkpointing); every engineering amendment (BatchNorm after the Cora smoke showed textbook BGRL collapse; ARPACK fail-fast; the neighbour cap) is timestamped before the rows it touched.

Pivotal verdicts (6 primary datasets):

comparison	mean AMI effect	95% CI	wins	verdict
gcn_gae vs mdt_select_all	0.068	[0.152, +0.017]	1/6	dead_end
sage_gae vs mdt_select_all	0.072	[0.160, +0.015]	1/6	dead_end
	0.286		1/6	dead_end

comparison	mean AMI effect	95% CI	wins	verdict
sage_bgrl vs mdt_select_all		[$\hat{\alpha}$ '0.492, $\hat{\alpha}$ '0.080]		
gcn_bgrl vs mdt_select_all	$\hat{\alpha}$ '0.103	[$\hat{\alpha}$ '0.236, +0.029]	2/6	inconclusive
(all four) vs mdt_consensus_all	$\hat{\alpha}$ '0.08 to $\hat{\alpha}$ '0.30	upper $\hat{\alpha}$ '0.03	$\hat{\alpha}$ '0.02 to 2/6	dead_end / inconcl.

Why the GNNs lose: their only wins are on the datasets whose graphs the screen flagged as weak (imdb +0.02 to +0.04, amazon_ratings +0.005 to +0.02) – the regime where MDT's diffusion also has nothing to work with. Where graphs are informative, spectral consensus extracts the signal at least as well as message passing, without training. Supporting reads: GNNs beat raw KMeans-on-features 5/6 (+0.04 to +0.08, not significant), so they are not broken – they are simply not better than the spectral route. Distilled MLPs match their GNN teachers (graph unnecessary at inference – replicating the first study's OOS finding at full scale). Controls behaved as designed: on roman_empire (anti-correlated graph) message passing hurts every GNN, and minesweeper (structure-only signal) is the one regime with a real GNN edge (+0.05 over everything spectral).

4.4 Enhancement round E1/E3 (120 rows) – using GND ideas inside MDT

Rationale: the grid located MDT's two failure modes – noise views poison uniform consensus, and diffusion cannot exploit heterophilous structure. Before licensing any learned (GNN) component, the pre-registered rule was heuristic-first: if the cheap version shows no effect, the learned version is not run.

- **E1, per-node view gating** (gate = cosine between a node's features and its view-v neighbour average): failed its primary readout ($\hat{\alpha}$ '0.085 vs consensus, 1/6) and failed the IMDB rescue. The failure is diagnostic, not random: the signal anchors on feature agreement, so across all 9 datasets it helps exactly where features are at least as informative as graphs (ACM: the sharpened variant reaches ACC 0.886 / NMI 0.685 – O2MAC-level, exploratory status) and destroys performance where graphs dominate (DBLP 0.55 to 0.03). A gate must estimate view quality, not feature agreement; that reframing is the round's real output.
- **E3, A² two-hop views**: dead as an enhancement ($\hat{\alpha}$ '0.033), and decisive in the other direction – it does **not** close the GNN's minesweeper edge ($\hat{\alpha}$ '0.048 vs gcn_gae there). So the one graph-specific GNN gain survives a spectral counter-attack; it is real, small, and lives on a synthetic control.
- **Validation on untouched amazon** (40 rows): the dataset turned out uninformative (no method above 0.05 AMI), so the exploratory ACM number stays exploratory. The gate's direction behaved exactly as the feature-anchoring model predicts even there (gates 0.011 to 0.014 > consensus 0.001, because amazon's graphs are pure noise and its features weakly informative).

5. Limitations of my own study (things a reviewer will attack)

Stated plainly because the first study earned criticism for not doing so:

1. **CPU-only budget compromises:** 150 epochs, top-64 neighbour cap, fixed lr/hidden, no per-method tuning. Defensible (tuning without labels is itself unsolved, and the selection criterion was identical for every arm), but a GNN advocate will argue tuned GNNs with more capacity could close the gap. The intervals cannot exclude a +0.02 GNN advantage on most comparisons â€” only large advantages are excluded.
2. **sage_bgml collapses on 7/9 datasets** â€” my hand-rolled BGRL is evidently sensitive; gcn_bgml is the only stable BGRL cell. GAE cells are solid. If BGRL matters to a reviewer, a reference implementation should replace mine.
3. **The MLP+SSL feature ceiling collapsed** (~0.02 AMI everywhere), so the "fusion value" comparisons lean on raw KMeans instead. A proper feature-SSL baseline (e.g. SimCLR-style with a validated recipe) is missing.
4. **The ledger promised spectral clustering on embeddings as a robustness check; it was never run.** KMeans is the single clustering protocol in all 600 rows.
5. **Placement protocol mismatch:** full-graph vs test-split NMI for the DMGI-lineage anchors; the comparison is honest about direction, not exact about magnitude.
6. **Two of the nine datasets (amazon, amazon_ratings) carry no recoverable signal for any method**, reducing effective statistical power below the nominal count.
7. The trajectory-selection port is lightweight (16 sampled + one-hot candidates), not the lab's exact select_trajectory.

None of these, in my assessment, threatens the headline direction â€” every comparison points the same way and the placement result is strong â€” but they bound how far the claims can be pushed in a paper.

6. What I consider established

1. Given identical inputs on intact graphs, **no tested GNN beats MDT consensus**; the burden of proof has moved to the GNN side.
2. **Zero-training MDT with graph views is competitive with published multiplex GNN clustering** (DBLP: above HDMI; ACM: 0.06 under O2MAC).
3. **Three lab tooling fixes** independent of any GNN question: modularity-based selection (silhouette is degeneracy-blind), cosine kNN for BoW-like features, consensus-over-trajectories preferred to single-trajectory selection (+0.05 to +0.10 AMI on acm/dblp).
4. **Per-node view weighting has real leverage** (exploratory ACM ACC 0.886) but needs a view-quality signal; feature-anchored gating is understood and closed.
5. The **only surviving graph-specific GNN gain** is +0.05 on a synthetic structure-only control, and a cheap spectral A^2 view does not replicate it.

7. Next plans, ranked by information per unit cost

1. **Decisions I need from you** (no compute): publication route â€” my recommendation is a new experiments section in the mdt_ddm paper (multiplex benchmarks, placement + grid) rather than a standalone negative-results paper; whether the GNN thread stays closed; whether the tooling fixes should be ported into the lab's MDT code now.

2. **$\hat{I}_{\pm}$ -normalized MDT views (~half a day, pre-registered).** The diffusion-maps $\hat{I}_{\pm}$ -family is implemented in `src/diffusionloss.py` but has never touched the MDT line: all MDT transitions are plain row-stochastic ($\hat{I}_{\pm}=0$), and the single-view DDM experiments fix $\hat{I}_{\pm}$ per dataset without any sweep. The noise views that poison consensus (ACM PLP, DBLP PATAP/PTP) are precisely hub-dominated, and $\hat{I}_{\pm}=1$ exists to remove degree/density bias before diffusion. One-line change to view construction; targets the measured failure mechanism directly; cheaper than any learned component. This is my top experimental candidate.
3. **Cross-view agreement gate (~1 week, new pre-registration).** Gate view v at node i by the agreement of its view- v neighbourhood with the other views' neighbourhoods (no anchoring on any single view). The licensed successor to E1; if its heuristic shows an effect, a small GNN gate becomes the first justified GNN component in the pipeline.
4. **Paper-hardening reruns (2~3 days):** protocol-aligned placement (test-split NMI for the DMGI lineage), spectral-clustering robustness check over the existing embeddings, and a reference BGRL to replace mine's items 2, 4, 5 of the limitations list.
5. **A real validation dataset for the sharp-gate result:** the O2MAC-lineage DBLP-4057 / IMDB-4780 files would double as a cross-lineage check of the placement claims.

8. Process note

Everything above is reproducible from the repo: pre-registrations and all amendments are in the ledger with their evidence; all 600 result rows are in `results/graph_mdt/*.jsonl`; verdict labels come only from `experiments/graph_mdt/verdicts.py`. The discipline cost real results's the sharpened gate cannot claim its best number because the smoke run touched the rescue dataset's and that is exactly what makes the rest of the numbers defensible.